SYSDESIGN REFERENCE

Design a File Storage Service

Requirements-to-architecture reference walkthrough.

Store large immutable or versioned objects separately from metadata and access policy.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / FILE-STORAGE-SERVICE

SYSDESIGN

Design a File Storage Service

CORE IDEA

Store large immutable or versioned objects separately from metadata and access policy.

Reference architecture decision chain

01	02	03	04
Multipart Upload	Object Keys	Metadata Db	Replication

Decisions to defend

+ Multipart upload
+ object keys
+ metadata DB
+ replication
+ checksums
+ and signed access.

Failure modes to test

+ Partial upload
+ orphaned chunks
+ corruption
+ unauthorized sharing
+ and hot objects.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a File Storage Service

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS
REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.